Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 45

Searching and Sorting

Covers Chartrand & Zhang, Section 6.4

Status of this set

Ungraded practice. Solutions released with it. Show each pass, not just the sorted list.

Part I: How to Practice This Material

The five algorithms at a glance

Algorithm	Complexity	Requires
Linear search	$O(n)$	nothing
Binary search	$O(\log n)$	a sorted list
Bubble sort	$\Theta(n^2)$	nothing
Insertion sort	$O(n^2)$	nothing
Merge sort	$\Theta(n \log n)$	nothing

Three mistakes to expect

1. **Running binary search on an unsorted list.** The halving step is meaningless without order.
2. **Reporting only the final sorted list.** The passes are the answer.
3. **Confusing the merge bound with the merge-sort bound.** Merging two lists costs at most $m + n - 1$; sorting costs $\Theta(n \log n)$.

Part II: Practice Problems

Section A — Searching

1. Use binary search to determine whether 27 appears in $s : 2, 5, 9, 14, 20, 23, 27, 31$. Show each pass.

2. Repeat for the target 25, which is not in the list. How does the algorithm know to stop?

Section B — Sorting

3. Sort $s : 9, 3, 7, 1$ by bubble sort, showing the list after each pass. How many comparisons does it make?

4. Merge the sorted lists $2, 4, 8$ and $1, 3, 5, 9$. How many comparisons does it take, and what does Theorem 6.43 allow?

5. Sort $s : 5, 2, 8, 1, 9, 3, 7, 4$ by merge sort. Show the divide and merge stages.

Section C — Writing

6. In five to eight complete sentences, explain why merge sort beats bubble sort by a factor that grows with n, and why no faster computer closes the gap.